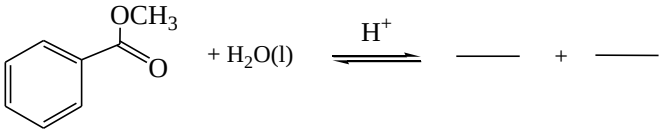
OROMIA IFA BORU SPECIAL BOARDING SCHOOL@2016

CHEMISTRY WORKSHEET FOR GRADE (9-11)

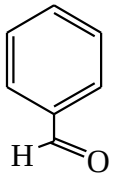
[ON HYDROCARBONS, IMPORTANT INORGANIC COMPOUNDS, CHEMICAL KINETICS, CHEMICAL EQUILIBRIUM AND OXYGEN-CONTAINING ORGANIC COMPOUNDS]

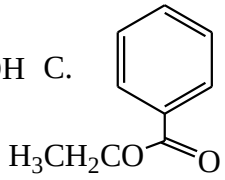
I. CHOOSE THE MOST APPROPRIATE ANSWER FROM THE SUGGESTED OPTIONS

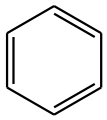
1. The process of removing coloring matter from wood pulp and increasing its brightness.
   A. Annealing
   B. Bleaching
   C. Soaking
   D. Scudding
2. Given the reaction: $SiCl_2 + 2Cl^- \longrightarrow [SiCl_4]^-$, $SiCl_2$ would be:
   A. Lewis base
   B. Electron pair donor
   C. Lewis acid
   D. It is an electron rich specie
3. Given the reaction:

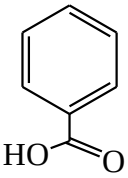


The product of the above reaction would be:

A.  + CH_3OH

C.  + CH_3CH_2OH

B.  + CH_3OH

D.  + CH_3OH

4. Given two different molecule (I) and (II); what is the product of the reaction below.



360 °C, Cu

---------- & ---------- respectively

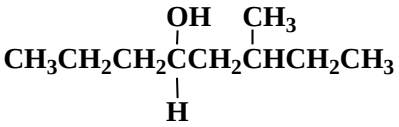
   A. CH_3COOH & CH_3CHO
   B. $CH_3CH_2COCH_3$ & CH_3CH_2CHO
   C. CH_3CH_2COOH & $CH_3CH_2CH_2CH_3$
   D. $CH_3CH_2CH_2COOCH_3$ &|$CH_3CH_2CH_2CHO$
5. Industrially; ethanol can be prepared by:
   A. Dehydrohalogenation of ethyl halide with potassium hydroxide (KOH).
   B. Reaction between calcium carbide with water.
   C. Fermentation of carbohydrates such as sugar.
   D. Reaction between sodium acetate and soda lime.

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6. What is the **IUPAC** name of the following organic molecule?

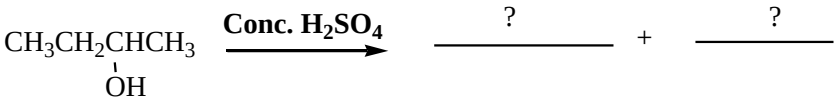


A. 3-methyl -4-hydroxy octane
C. 6-methyl-4- hydrogen-4-hydroxy octane
B. 2-ethyl-4-octanol
D. 6-methyl-4-octanol

7. Given the reaction: NH_3 (ι) + H^+ (aq) ⟶ NH_4^+ (aq). Which of the following is true?
A. H^+ is electron loving species/electrophile
C. NH_3 is Lewis's acid
B. In NH_3 molecule, two lone pair es^- exist on the 3H's
D. H^+ is Lewis's base

8. Given that: CO, CO_2 & I_2. Which arrangement of bond strength is correct for the covalent molecules:
A. $CO < CO_2 < I_2$
C. $CO_2 < I_2 < CO$
B. $CO < I_2 < CO_2$
D. $I_2 < CO_2 < CO$

9. What is the product of the reaction:



A. $CH_3CH_2CH{=}CH_2$ & CO_2
C. $CH_3CH_2CH{=}CH_2$ & H_2O
B. $CH_3CH{=}CHCH_3$ & H_2O
D. $CH_3\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2CH_3$ & H_2O

10. Which one of the following molecules belongs to ketones group?
A. $CH_3\,(CH_2)_5CO\,(CH_2)_2\,CH_3$
C. $CH_3\,CH_2O\,CH_2\,CH_3$
B. $CH_3(CH_2)_5\,CHO$
D. $CH_3\,(CH_2)_5\,CH(OH)CH_2CH_2CH_3$

11. The most common reaction in which benzene undergo is said to be:
A. Combustion reaction
B. Elimination rxn.
C. Substitution rxn.
D. Synthesis rxn.

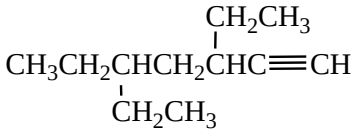
12. The chemical used to pests that damage plants growth, development & crop yields is said to be:
A. Fertilizers
B. Herbicides
C. Sedatives
D. Pesticides

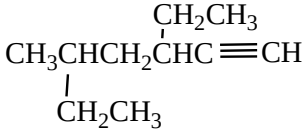
13. Which of the following hydrocarbons does belongs to the homologous series of alkenes?
A. C_3H_4
B. C_8H_{16}
C. C_6H_{10}
D. C_9H_{16}

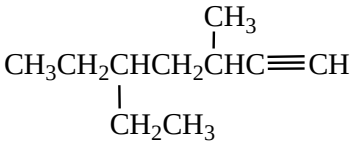
14. Silver is an element with atomic number 47 is found in d-block in the periodic table. What compound would you expect from combination of Ag (I) and SO_4^{2-}?
A. $AgSO_4$
B. $Ag\,(SO_4)_2$
C. Ag_2SO_4
D. $Ag_3(SO_4)_2$

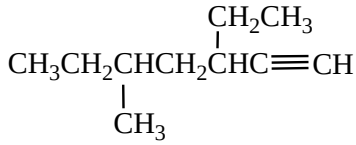
15. Which arrangement is correctly represents descending order the oxidation number of the central atoms in the following ternary acids?
A. $HClO_4 < HClO_3 < HClO < HClO_2$
C. $HClO < HClO_2 < HClO_3 < HClO_4$
B. $HClO_3 < HClO < HClO_2 < HClO_4$
D. $HClO_4 < HClO_3 < HClO_2 < HClO$

16. Which structure correctly have an IUPAC name 3,5-diethyl hept-1-yne?

A. 

C. 

B. 

D. 

17. A Bronsted Lowry acid is defined as a substance that:
   A. Furnishes hydronium ion in an aqueous solution
   B. Accept a pair of electrons from another species
   C. Accept proton from another species
   D. Donate proton for another species

18. What is the function of adding gypsum salt to the cements' raw materials during industrial manufacture of cements?
   A. In order to solidify cement
   B. In order to make cement become fine powder
   C. To acidify cement
   D. To enhance impurities in cement

19. In the Arrhenius equation $\mathbf{k = Ae^{-Ea/RT}}$, at what condition the rate constant **k** equal to the Arrhenius frequency factor, **A**?
   A. When the temperature becomes infinite
   B. When the fraction of the molecules crossing the energy barrier is unity
   C. When 50% reactants are converted to product
   D. When activation energy is not needed to reactants for undergoing chemical change

20. The temperature dependence of rate constant (k) of a chemical reaction is written in terms of Arrhenius equation, $\mathbf{k = Ae^{-Ea/RT}}$. Activation energy (Ea) of the reaction can be calculated by plotting:
   A. logk vs T B. logk vs 1/T C. k vs 1/logT D. k vs T

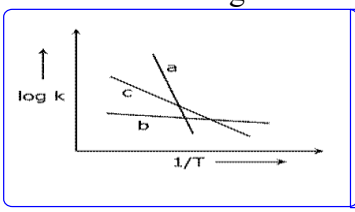
21. The temperature dependence of the rate of a chemical reaction is given by Arrhenius equation, $\mathbf{k = Ae^{-Ea/RT}}$. Which of the following graphs will provide a straight line?
   A. lnA vs 1/T B. lnA vs E_a C. lnk vs 1/T D. lnk vs $-Ea/R$

22. Rate constant k of a reaction varies with temperature according to the equation $\mathbf{logk = constant - E_a/2.303R \times (1/T)}$ where E_a is the energy of activation for the reaction. When a graph is plotted for **logk vs 1/T** a straight line with a **slope −6670k** is obtained. The activation energy for this reaction will be: ($R = 8.314$ J. $K^{-1}mol^{-1}$)
   A. 122.65 $kJmol^{-1}$ B. 127.71 $kJmol^{-1}$ C. 150.00 $kJmol^{-1}$ D. 142.34 $kJmol^{-1}$

23. Which of the following factor will **lower** the activation energy for a reaction?
   i. Adding a suitable catalyst
   ii. Increasing the concentrations of reactants
   iii. Raising the temperature of the reaction
   iv. All of these options will lower the activation energy for a reaction

24. Activation energy is required to initiate:
   A. Neither exothermic or endothermic reactions
   B. Both exothermic and endothermic reactions
   C. Endothermic reactions only
   D. Exothermic reactions only

25. Consider the following plots of **log(k)** versus **1/T** for three different reactions.
Which of the following reaction has highest activation energy (Ea)? (where, **k** is the rate constant).



A. Eb
B Ea
C. Ec
D. All have equal activation energies.

26. According to the Arrhenius equation, $\mathbf{k = Ae^{-EA/(RT),}}$ what happens to the rate constant **k** of the reaction as the temperature of the reaction is increased?
A. The rate constant would increase
B. There is no effect in either direction
C. It cannot be determined from the given information
D. The rate constant would decrease

27. According to Arrhenius equation, the rate constant (k) is related to temperature (T) as:
A. $\ln(k_2/k_1) = Ea/R(1/T_1 - 1/T_2)$ C. $\ln(k_2/k_1) = Ea/R(1/T_1 + 1/T_2)$
B. $\ln(k_2/k_1) = -Ea/R(1/T_1 - 1/T_2)$ D. $\ln(k_2/k_1) = -EaR(1/T_1 + 1/T_2)$

28. Consider the reaction which ammonia is synthesized from nitrogen and hydrogen gases:
How is the rate of formation of ammonia related to the rate of consumption of hydrogen?
A. The rate of formation of ammonia is two thirds the rate of consumption of hydrogen
B. The rate of formation of ammonia is half the rate of consumption of hydrogen
C. The rate of formation of ammonia is equal to the rate of consumption of hydrogen
D. The rate of formation of ammonia is 1/5 times the rate of consumption of hydrogen

29. If $\mathbf{K_1}$ and $\mathbf{K_2}$ are the equilibrium constants at temperature T_1 and T_2, where $\mathbf{T_2 > T_1}$, then:
A. $K_2 = K_1$ when E_a is 0 C. $K_2 > K_1$ when E_a is +ve
B. $K_2 < K_1$ when E_a is -ve D. $K_2 < K_1$ when E_a is +ve

30. In the reaction, $2A + B \longrightarrow 2C + D$, the $-\Delta[A]$ per time was found to be 5.0. What is the rate of change of $\Delta[B]$?
A. 25 M/min B. 10 M/min C. 2.5 M/min D. 5.0 M/min

31. Given the reaction: $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$. The rate of the forward reaction increases by adding more SO_2 because the
A. Number of molecular collisions will increase C. Forward reaction is endothermic
B. Temperature will increase D. Reaction will shift to the left

32. For the reaction $A \longrightarrow B$, the activation energy is 125 kJ/mol and the heat of reaction, ΔH = 50.00 kJ/mol. What is the for the reverse reaction in kJ/mol?
A. 125 kJ/mol B. -75 kJ/mol C. 75 kJ/mol D. -125 kJ/mol

33. When NH_4NO_3 is dissolved in water, the temperature of the water decreases. When NaOH is dissolved in a separate water sample, the temperature of the water increases. Based on these observations, it can be concluded that the dissolving of:
    A. Both salts are endothermic.
    B. Both salts are exothermic.
    C. NH_4NO_3 is exothermic and the dissolving of NaOH is endothermic.
    D. NH_4NO_3 is endothermic and the dissolving of NaOH is exothermic.
34. Which of the following is an example of a heterogeneous reaction?
    A. Combustion of methane C. Dissolution of sugar in water
    B. Formation of rust on iron D. Decomposition of hydrogen peroxide
35. For a reaction to occur, reactant molecules must:
    A. Collide with sufficient energy and proper orientation
    B. Collide with minimal energy and any orientation
    C. Collide with maximal energy and proper orientation
    D. Collide with any energy and random orientation
36. The term 'collision theory' is used to explain:
    A. How catalysts work in reactions C. How particles interact during reactions
    B. The role of temperature in reactions D. The equilibrium state of reactions
37. When the two reactants are in the same phase, then which common phase will ensure that the products are formed at the fastest pace?
    A. Solid B. Liquid C. Gas D. The rate of a rxn. is independent of the phase of the $\mathbf{R}_s$.
38. Which of the following is not a direct factor affecting the rate of a reaction?
    A. Temperature B. Presence of catalyst C. Order of reaction D. Molecularity
39. Which of the following will lead to an increase in the rate of the reaction?
    A. Decrease in temperature C. Addition of catalyst
    B. Decreasing concentration of reactants D. Addition of inhibitor
40. The rate of a chemical reaction tells us about: -
    A. The reactants taking part in the reaction C. The products formed in the reaction
    B. How slow or fast the reaction is taking place D. It does not tell anything
41. The average rate and instantaneous rate of a reaction are equal
    A. At the start C. At the end
    B. In the middle D. When two rates have a time interval equal to zero
42. If the reaction quotient **Q** has a smaller value than the related equilibrium constant, **K** then:
    A. The reaction is not at equilibrium, and will make more products at the expense of reactants.
    B. The reaction is not at equilibrium, and will make more reactants at the expense of products.
    C. The value of K will decrease until it is equal to **Q**.
    D. The value of K will ascend until it is equal to **Q**.

43. Consider this equilibrium reaction in a sealed container. $H_2O\ (g) \rightleftharpoons H_2O\ (l)$, which one of the following will be the effect on the equilibrium of increasing the temperature from 20ºC to 30ºC?
   A. More of the water will be in the gaseous state of at equilibrium.
   B. More of the water will be in the liquid state at equilibrium.
   C. At equilibrium the rate of condensation will be greater than the rate of evaporation.
   D. At equilibrium the rate of evaporation will be greater than the rate of condensation.

44. For the reaction: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, at a certain temperature the equilibrium concentrations in mol/liter are $[H_2] = 0.2$, $[I_2] = 0.2$ and $[HI] = 2.0$, then based on this given data which one of the following is the equilibrium constant($\mathbf{K_c}$)?
   A. 0.002 B.10 C. 20 D.100

45. Which statement is correct about the effect of adding a catalyst to a system at equilibrium?
   I. The rate the of forward reaction increases.
   II. The rate of reverse reaction increases.
   III. The yield of the products increases.
   A. I only B. I and II only C. III only D. I, II and III

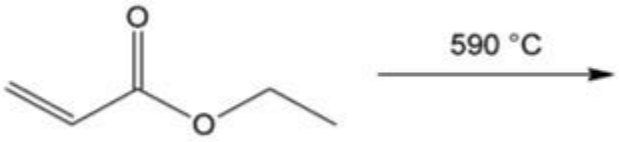
46. For the reaction: $N_2O_4\ (g) \rightleftharpoons 2NO_2\ (g)$, $\mathbf{K_c} = 8 \times 10^{-4}$ mol/cm, in an equilibrium of mixture these two gases $[N_2O_4] = 8 \times 10^{-2}$ mol/cm^3, then which one is the equilibrium concentration NO_2 in mol/cm^3?
   A. 8.0×10^{-1} B. 8.0×10^{-3} C. 8.0×10^{-2} D.4.5×10^{-4}

47. For the reaction: $\mathbf{C_2H_4\ (g) + H_2\ (g) \rightarrow\ C_2H_6\ (g)}$, $\Delta H = -137KJ$, which statement about the information is correct?
   A. The total energy of the bonds broken in the reactant is greater than the total energy of the bonds formed in the product.
   B. The bonds broken and the bonds made are of the same strength.
   C. The total energy of the bonds broken in the reactants is less than the total energy of the bonds formed in the product.
   D. No conclusion can be made about the sums of the bond enthalpies in the product compared with reactants.

48. Under what conditions can we absolutely say a system is at equilibrium at constant temperature and pressure?
   A. Change in Enthalpy is equal to Zero C. Change in Gibbs free energy is equal to Zero
   B. Change in Entropy is greater than Zero D. Change in Enthalpy is greater than Zero

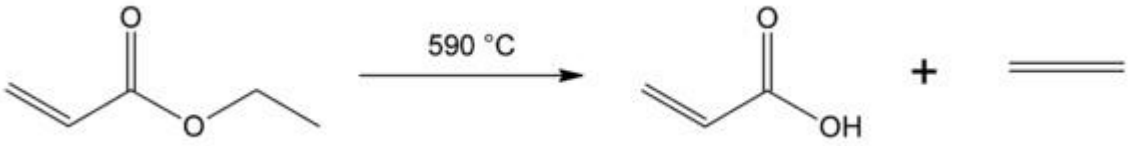
49. In the following reaction, what is equal to the rate of production of NO gas?
   i.e. $4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$
   A. The rate of production of NH_3 gas C. One third the rate of production of water
   B. Four fifths the rate of disappearance of O_2 gas D. One quarter the rate of disappearance of NH_3

50. Each of the following is true about heterogeneous catalyst EXCEPT:
    A. It does not undergo a permanent change
    B. It is in the same phase as the reacting particles
    C. Its presence lowers the activation energy of the overall reaction
    D. Its presence decreases the potential energy of the activated complex
51. In general, reaction rates double when the temperature is increased by 10°C. The temperature of a reaction is increased by 40°C. The rate of the reaction will increase by a factor of:
    A. 2 B. 4 C. 8 D. 16
52. Which of the following elementary reactions is a termolecular reaction?
    A. $A + 2B + C \rightarrow D$ B. $A + B + B \rightarrow C$ C. $2A + 2B + 2C \rightarrow 2D$ D. $2A + B + C \rightarrow D$
53. Which of the following explains the increase in the reaction rate by a catalyst?
    A. Catalyst decreases the rate of backward reaction so that rate of forward reaction increases
    B. Catalyst provides extra energy to reacting molecules so that they produce effective collisions
    C. Catalyst provides an alternative path of lower activation energy to the reactants
    D. Catalyst increases the number of collisions between the reacting molecules
54. What is the characteristic smell for ester?
    A. Fruity like smell B. Fish like smell C. Rotten egg smell D. Alcoholic smell
    Explanation: Many esters have distinctive fruit-like odors, and many occur naturally in the essential oils of plants. This has also led to their commonplace use in artificial flavorings and fragrances when those odors aim to be mimicked.
55. What will be the product of given reaction?



    A. Ether and alcohol C. Alcohol and sodium carboxylate
    B. Aldehyde and alcohol **D.** Carboxylic acid and alkene

    Explanation: Ester pyrolysis in is a vacuum pyrolysis reaction converting esters containing a β-hydrogen atom into the corresponding carboxylic acid and the alkene.

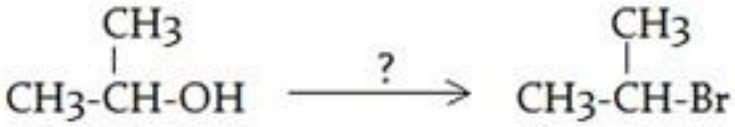


56. What is the order or solubility in water of the following compound?
    i) ethyl methanoate ii. ethyl butanoate iii. ethyl ethanoate iv. ethyl propanoate
    A. i > ii > iii > iv **B**. i > iii > iv > ii C. i > ii > iv > iii D. iii > i > iv > ii
    Explanation: As chain length increases, the hydrocarbon portion forces itself between water molecules, breaking the relatively strong hydrogen bonds between water molecules without offering an energetic compensation; furthermore, the water molecules are forced into an ordered alignment along the chain, decreasing the entropy in the system. This makes the process thermodynamically less favorable, and so solubility decreases.

57. Among the following, which is least acidic?
A. Phenol **B**. O-cresol C. p-nitrophenol D. p-chlorophenol
Explanation: O-cresol is least acidic because Presence of electron withdrawing groups at ortho and para position increases the acidic strength, whereas electron releasing or donating groups at ortho and para position decreases acidic strength. In cresol (ortho or para cresol), methyl group is +I group, so it destabilizes the anion formed after ionization, hence cresol is less acidic than carbolic acid (phenol).

58. Which of the following alcohols would be most soluble in water?
A. Propanol B. Hexanol C. Pentanol D. Butanol
Explanation: Propanol is the most soluble in water among given alcohols, as solubility of alcohols in water decreases with increase in molecular mass.

59. What is the best reagent to convert isopropyl alcohol to isopropyl bromide?



A. HBr **B.** $SOBr_2$ C. Br_2 D. CH_3MgBr
Explanation: Thionyl bromide will react with secondary alcohols and give a reasonable percent yield of the bromide.

60. The major reason that phenol is a better Bronsted acid than cyclohexanol is:-
A. It is a better proton donor
B. The cyclohexyl group is an electron donating group by induction, which destabilizes the anion formed in the reaction
C. Phenol is able to stabilize the anion formed in the reaction by resonance
D. The phenyl group is an electron withdrawing group by induction, which stabilizes the anion formed in the reaction.
Explanation: Whenever resonance exists, it usually is a very important factor to be considered in decisions about structure and reactivity correlations and in many cases, it is the most important factor. In this case there is delocalization of the electrons into the benzene ring.

61. Which one of the following compd. is obtained by the oxidation of **1° alcohol** with nascent oxygen?
A. Alkanal B. Alkanone C. Ether D. Amine
Explanation: Oxidation of alcohols to aldehydes is partial oxidation; aldehydes are further oxidized to carboxylic acids. Conditions required for making aldehydes are heat and distillation. In aldehyde formation, the temperature of the reaction should be kept above the boiling point of the aldehyde and below the boiling point of the alcohol.

62. The reaction of carboxylic acids with alcohols catalyzed by conc. H_2SO_4 is known as: -
A. Dehydration B. Saponification **C.** Esterification D. Neutralization
Explanation: When a carboxylic acid is treated with an alcohol and an acid catalyst, an ester is formed (along with water). This reaction is called the esterification.

63. Which one is the correct order of reactivity of different types of alcohol towards hydrogen halide?
A. 1° alcohol > 2° alcohol > 3° alcohol C. 2° alcohol > 1° alcohol > 3° alcohol
B. 3° alcohol > 1° alcohol > 2° alcohol **D.** 3° alcohol > 2° alcohol > 1° alcohol

Explanation: Reaction of HX with alcohol depends on the degree of carbonation generated due to heterocyclic cleavage of HX bond. If alcohol is primary alcohol then reaction will proceed through Sn2 for tertiary alcohol it will proceed with Sn1. For secondary alcohol it can go through both path (SN^1 or SN^2). SN^1 being ionic in nature takes place very fast while SN^2 is molecular in nature reacts slowly. Hence, we can sum up above, Rate of reaction is 3° alcohol > 2° alcohol > 1° alcohol.

64. Which of the following is the most suitable requirement for the dehydration of ethanol to form an ether?
A. Sulphuric acid at 413K C. Sulphuric acid at 443K
B. Sodium hydroxide at 413K D. Sodium hydroxide at 443K

Explanation: Alcohols undergo dehydration in the presence of protic acids to form either alkenes or ethers depending upon the reaction temperature conditions. At 413K, ethanol forms an ether with H_2SO_4.

65. Which of the following is not favourable for the proper dehydration of alcohol to form ether?
A. Good temperature control during reaction C. Excess of alcohol
B. Presence of protic acids **D.** Presence of bulky alkyl groups in the alcohol

Explanation: Secondary and tertiary alcohols give alkenes as major products during dehydration, because elimination competes over substitution. Hence, this method is mainly used for primary alcohols.

66. What is the major product formed when ethanol is dehydrated with concentrated H_2SO_4 at 413K?
A. Ethene B. Methoxymethane C. Methoxyethane **D.** Ethoxyethane

Explanation: Symmetrical ethers are formed when alcohols are dehydrated with H_2SO_4 at controlled temperatures. This is an S_N2 reaction involving the attack of alcohol molecule on a protonated alcohol to give ethers.

67. Sodium methoxide on heating with bromoethane gives:
A. methoxymethane **B.** methoxyethane C. ethoxyethane D. diethyl ether

Explanation: This is an example of Williamson synthesis of unsymmetrical ether, where CH_3ONa is reacted with CH_3CH_2Br to form $CH_3CH_2OCH_3$, which is ethyl methyl ether.

68. The reaction between tert-Butyl chloride and sodium ethoxide gives:
A. tert-Butyl ethyl ether B. tert-Butyl methyl ether **C.** 2-Methylprop-1-ene D. butene

Explanation: The reactant sodium ethoxide is a strong nucleophile as well as a strong base, and hence elimination predominates over S_N2 to form alkene as a major product.

69. Which of the following is the least reactive functional group?
A. Alcohols **B.** Ethers C. Aldehydes D. Ketones

Explanation: This is because the (-O-) group in ethers does not contain any active site as compared to for example, hydroxyl group (OH) in alcohols. However, they undergo C-O bond cleavage in drastic conditions.

70. Catalytic reduction (Hydrogenation) of carbon monooxide with H_2 under high pressure and temperature gives methyl alcohol. What is this process is known as?
A. Baeyer's test B. Hofmann's reaction C. Kolb process **D.** BASF process

Explanation: Currently, a $Cu/ZnO/Al_2O_3$ catalyst system has been employed industrially in MeOH synthesis for over half a century (a worldwide demand of 75 Mtons•year−1). It was run mostly at elevated pressures. (5–10 MPa) and high temperature (493–573 K) by feeding a CO (5%)-CO_2 (5%) H_2 syngas stream.

OROMIA IFA BORU SPECIAL BOARDING SCHOOL@2016

CHEMISTRY WORKSHEET FOR GRADE (9-11)

[ON HYDROCARBONS, IMPORTANT INORGANIC COMPOUNDS, CHEMICAL KINETICS, CHEMICAL EQUILIBRIUM AND OXYGEN-CONTAINING ORGANIC COMPOUNDS]

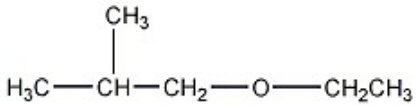
71. Which of the following statements is incorrect regarding preparation of alcohols?
    A. Aldehydes and Ketones on catalytic reduction by [H] give primary alcohol and secondary alcohol respectively
    B. Ketones on reaction with Grignard's reagent gives tertiary alcohol
    C. Aldehydes on reaction with Grignard's reagent gives primary or secondary alcohol
    D. Alkyl halides on reaction alcoholic potash gives alcohol

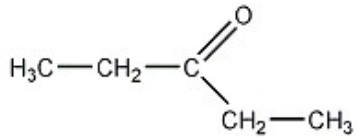
    Explanation: Whenever an alkyl halide reacts with alcoholic potash (or alcoholic sodium hydroxide), beta-elimination of the halide occurs, leading to the formation of an alkene.

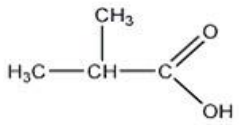
72. Which of the following steps is not used in the fermentation of starch into ethyl alcohol?
    A. Hydrolysis of starch into maltose by diastase present in barley
    B. Conversion of maltose into glucose by maltase present in yeast
    C. Conversion of glucose into ethanol by zymase present in yeast
    D. Conversion of sucrose into glucose and fructose by sucrase present in yeast

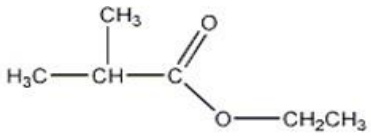
    Explanation: Conversion of sucrose into glucose and fructose by sucrase present in yeast steps is not used in the fermentation of starch into ethyl alcohol.

73. Which of the following compounds is an ester?

    A. 

    B. 

    C. 

    D. 

74. Which of the following is used as catalyst for the esterification of carboxylic acid and alcohol?
    A. Nitrous acid **B**. Sulphuric acid C. Sulphurous acid D. Nitric acid

    Explanation: The classic synthesis is the Fischer esterification, which involves treating a carboxylic acid with an alcohol in the presence of a dehydrating agent: $RCO_2H + R'OH \rightleftharpoons RCO_2R' + H_2O$ The equilibrium constant for such reactions is about 5 for typical esters, e.g., ethyl acetate. The reaction is slow in the absence of a catalyst. Sulfuric acid is a typical catalyst for this reaction.

75. What will be the product for the given reaction? **$CH_3OH + CO \rightarrow$?**
    A. Ethyl formate **B.** Methyl formate C. Ethyl acetate D. Methyl acetate

    Explanation: The carbonylation of methanol yields methyl formate, which is the main commercial source of formic acid. $CH_3OH + CO \rightarrow CH_3OCO\text{-}H$